

Augmented Reality Indoor Navigation with Smart Appliance control

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Abstract—This proposed work explores the application of augmented reality (AR) for indoor navigation in emergency situations and large spaces using Unity NavMesh and the A* algorithm. It provides real-time information about college classrooms and enhances accessibility for people with disabilities. The proposed system utilizes Vuforia and Unity to construct the AR environment, integrating QR code scanning using the ZXING SDK, as well as the smart control of light bulbs, fans, and other appliances through ESP32, relay modules, and the Blynk API. Experimental results demonstrate the effectiveness of our approach in improving navigation accuracy and accessibility for users. Furthermore, the system's real-time updates, user-friendly interface, and remote appliance control offer a versatile solution for various indoor environments. The proposed work presents a detailed explanation of the literature review, the operating architecture, discusses its social impact, limitations, and future scope, highlighting its potential to revolutionize indoor navigation and accessibility for all.

Keywords- *Augmented Reality, Indoor Navigation, Unity NavMesh, A* Algorithm, Accessibility, QR Code Scanning, ESP32, Blynk API, Experimental Results.*

I. INTRODUCTION

In various settings, including educational institutions, hospitals, shopping centers, and business buildings, indoor navigation presents significant challenges. Finding the quickest route to safety in an emergency is essential, yet effectively maneuvering in large areas can be challenging. Traditional navigation systems often fall short in providing real-time information updates and meeting the needs of people with impairments. This paper proposes an indoor navigation system based on augmented reality that includes real-time data, remote appliance control, and enhanced accessibility features to address these limitations.

Navigating complex indoor spaces has long been a challenge. Traditional methods rely on static signs and limited visual cues, leaving users to navigate through unfamiliar environments. This work aims to revolutionize indoor navigation by leveraging augmented reality. By overlaying real-

time guidance and information directly onto users' field of vision, our system empowers individuals to confidently and efficiently navigate indoor spaces, even in high-stress emergency situations. Through the utilization of Unity NavMesh and the A* algorithm, users can access dynamic route suggestions, ensuring they reach their destinations.

In conjunction with indoor navigation, our proposed system seamlessly integrates with the Internet of Things (IoT). By connecting to smart appliances within indoor spaces, users gain unparalleled control over their surroundings. Through ESP32, relay modules, and the Blynk API, our system allows users to remotely manage lighting, climate control, and other devices, optimizing energy usage, enhancing comfort, and ensuring a personalized indoor environment.

Key Contributions:

- **Enhanced Indoor Navigation:** The paper introduces an augmented reality-based indoor navigation system that utilizes Unity NavMesh and the A* algorithm. This system significantly improves navigation accuracy, providing real-time guidance and information to users in emergency situations and large spaces, addressing the challenges of traditional navigation methods.
- **Accessibility and IoT Integration:** The proposed system goes beyond navigation by seamlessly integrating with the Internet of Things (IoT). Users can remotely control lighting, climate, and other appliances through ESP32, relay modules, and the Blynk API, enhancing accessibility for all individuals and offering personalized indoor environments.
- **Social Impact and Future Potential:** The system's contributions extend to public safety, inclusivity for people with disabilities, and energy efficiency. It has the potential to revolutionize indoor navigation and transform indoor spaces into responsive environments. Future enhancements could involve wearable devices, integration with emergency services, and further customization options for users.

II. LITERATURE REVIEW

Teng-Wei Hew, Seow Hui Saw, Manoranjitham Muniandy and Jasmina Khaw [1] propose a visual-inertial method to establish a connection between the actual and virtual environments. The approach combines image processing knowledge with information from the device's motion sensing hardware to analyze images that were taken by the camera. Additionally, SLAM, an instant tracking system that enables the examination of the surrounding real environment, was applied. The approaches used were scene, feature point generation, virtual plane simulation, and object placement. However, the system had issues because there were no Android builds and only iOS users could utilize it.

Abuhashish and H. Kolivand [2] survey the use of augmented reality (AR) technology for pathfinding. AR is a technology that superimposes computer-generated images on a user's view of the real world, providing a way to interact with the environment in a more immersive way. The system uses a database of GPS coordinates to store the locations of interest in the environment. The user can select a location of interest by pointing the camera at the desired location or by providing the location radius. The system then calculates the path to the selected location based on the user's current location and the GPS coordinates of the selected location. Pathfinding is the process of finding a path between two points in a given environment. The paper discusses the advantages of using AR for pathfinding, such as its ability to provide users with real-time information about their surroundings, its ability to create more engaging and interactive experiences, and its potential to improve the efficiency of pathfinding. It also discusses the challenges of using AR for pathfinding, such as the need for accurate and reliable tracking, the need for lightweight and power-efficient devices, and the need for user-friendly interfaces.

Khan, Rehman Ullah [3] present a system for indoor navigation using annotated maps in mobile augmented reality (AR). The system uses a digital map that combines the standard paper map with annotations that provide coordinate information and other location details. The mapping is done using AR Core. The device overlays 3D maps and location data, such as the corresponding navigation path, 3D objects, and text-based visual information, onto the AR camera feed. As the user moves through the environment, the camera is estimated and a 3D model is reconstructed. The system is evaluated in a simulated environment and is found to be effective in providing indoor navigation assistance to users. The paper makes a number of contributions to the field of indoor navigation using AR. First, it proposes a new system that uses annotated maps. Second, the system is implemented on a mobile device, which makes it portable and easy to use. Third, the system is evaluated in a simulated environment, which provides a reliable way to assess its performance.

Mayuri Tamhane, Prathamesh Sarjekar, Chirag Gupta and Prof. V M. Kharche [4] propose a method for campus navigation using augmented reality (AR). The method uses the

Android SDK, SDK, and Eclipse IDE. It makes use of the Mobile Augmented Reality (MAR) concept. MAR is created by combining AR with wireless connectivity, location-based computing services (LBS), and responsive environments. The position is determined using the compass, gyroscopes, accelerometers, and GPS data. A network connection is required for this operation whenever it is used. The system is evaluated in a university campus environment and is found to be effective in providing navigation assistance to users. The paper also discusses the benefits and challenges of using AR for campus navigation.

A. Natarajan, A. Saideep and K. S. Reddy [5] propose a system for home automation using artificial intelligence (AI) and augmented reality (AR). The system uses a Raspberry Pi as the main computing unit. The Raspberry Pi is connected to four-channel relay, two bulbs, and two fans. The system uses image processing and a convolutional neural network to detect objects in the environment and control the appliances accordingly. These four equipment are bunched together and connected to a switch. And Image processing on digital images with the help of Convolution Neural Network is used to capture image and process, recognize and match to the database to prompt options specific for that object. Raspberry pi comes with a limited memory and as its specifications are upgraded the cost increases significantly. Raspberry Pi is only compatible with Linux OS and single point of failure of the Raspberry Pi will disrupt the entire signal. The sensor and megapixel quality of the device used to capture the image can affect the decisions made by the software. Identical appliances become difficult to distinguish as processing will be targeted to limited object area. Analyzing an image to detect an object is a heavy task in the sense of computation and memory as well.

M. A. Ankireddy, A. Rajath, M. Ruthwik Ganesh and M. Anuradha [6] discuss a system for augmented reality (AR) rendered for Internet of Things (IoT) applications. The system uses smart glasses embedded with a Raspberry Pi Zero, a Pi camera, and touch sensors. The smart glasses are aware of the environment and can detect targets. When a target is detected, the relevant information is sent to a server. The input to the Vuforia engine for target detection is given in the form of a continuous video footage recorded by the Pi camera embedded on the smart glasses. The reference image stored in the database to match with the appliance has to be extracted from the real-time video stream which goes on continuously, causing overhead. To address this, an additional Pi camera is used that can be replaced by the mobile phone's inbuilt camera. Brute-Force Matching which computes a hamming distance between two sets and returns the number of matches is used to extract the reference image from the video stream. The system is evaluated in a simulated environment and is found to be effective in detecting targets and rendering AR content. However, the paper also discusses some of the challenges of using the system. Some of the challenges faced by the system are: The system is computationally expensive. It requires a good internet connection. The system is not yet

robust enough to handle occlusions and other challenges in real-world environments.

III. PROPOSED SYSTEM OPERATING ARCHITECTURE

Our system integrates multiple modules working in harmony to create an optimal solution. Module-wise implementation is

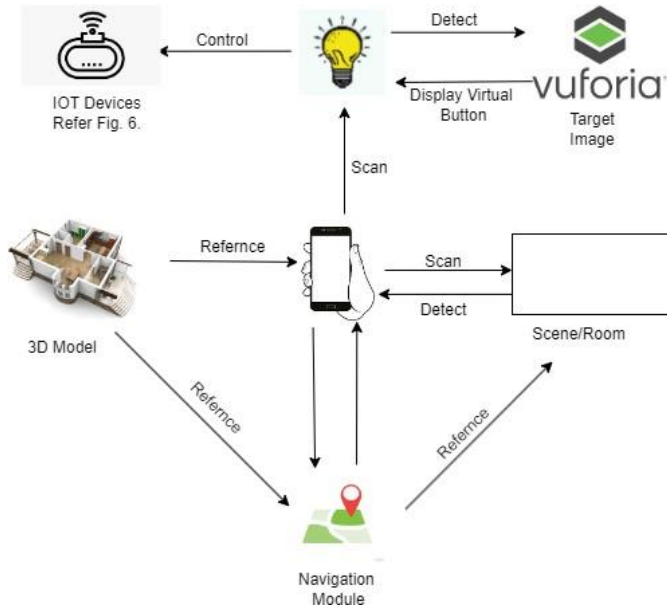


Fig. 1. Proposed System Architecture

discussed further:

A. 3D Modelling

First with the reference blueprint of the chosen area and detailed measurement of each obstacle like walls, a 2d model was built using SmartDraw. It is a versatile software tool that allows users to create 2D models of floors efficiently. SmartDraw provides an extensive library of pre-designed shapes that can be easily resized, rotated, and aligned to accurately depict the floor layout. SmartDraw automatically adjusts the scale of the floor plan, ensuring accurate measurements and proportions.



Fig. 2. 2D model of the chosen area

The 2D model was exported in a compatible file format, such as a vector-based format like SVG. In Unity, the exported 2D floor plan can be imported as a texture or image onto a plane or surface. This will provide the foundation for the 3D model creation. The plane can be positioned and scaled to

match the dimensions and layout of the floor plan. 3D elements in Unity, such as walls, furniture, and objects, using built-in modeling tools or external 3D software were added to replicated the real world indoor space. This model acts as an input to the navigation system.

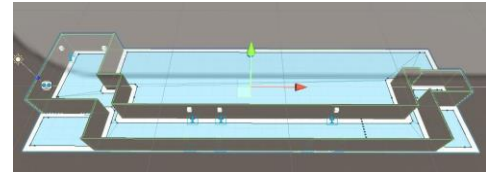


Fig. 3. 3D model of the chosen area

B. Indoor Navigation using Unity NavMesh A*

• A* Algorithm

It is a "Dijkstra" based search algorithm. The purpose of the heuristic search is to evaluate all the details in the state space, choose the best from the starting point, and begin the node-to-node search until the target node is found. The location of the test is important in heuristic search, and using different tests will lead to different results. The evaluation function is heuristic and represents the estimated cost of migrating from the current node to the target node. We often use the Manhattan estimation function to estimate costs in search and search problems. The following model represents the performance evaluation of the current node:

$$F(n) = H(n) + G(n) \quad (1)$$

$F(n)$ is an evaluation, $H(n)$ is an estimate of the shortest path from n to a point, $G(n)$ is the shortest path from one of n to the starting point, where: $1 \leq n$. [7] To record the location information of moving objects while navigating, the A* algorithm searches the map and uses the location of the objects to generate a match. Before using the A* algorithm, the map must be divided into sections and each section should act as a node. The simplest division is a quadrilateral since various terrains on the scene can be divided into different shapes like a four-forked tree or a convex polygon map.

• Unity NavMesh

The navigation system "Navmesh grid" is built in Unity 3D; All that is needed is to configure the map and add the "Navmesh Agent" component to the user. In Unity 3D, objects are the building blocks of graphics. The mesh component can be used to create the shape of the object, the force component can be used to control the position of the object, and the component can be used to create the properties of the object.[8] Like the A* algorithm, "Navmesh" needs to split the map into different types. However, if the polygons are regular, the A* algorithm works fine, which is not necessary for the latter. Light irradiation is the most used algorithm for "Navmesh" pathfinding. As shown in Figure 3, the

navigation mesh consists of random convex polygons. Gray grids represent inaccessible areas, and white grids represent accessible areas. All polygon vertices in the mesh are stored clockwise.

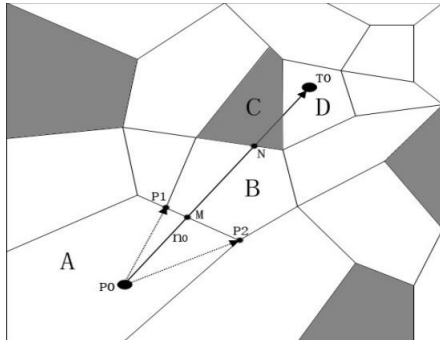


Fig. 4. Light irradiation

First we need to determine whether P_0 and the polygon intersect or are parallel and the rays entering the second. Check the connection between polygon A and $r = (p_0, n_0)$ using the example. Cross the a along each side e of the polygon to obtain the start and end points of each side.

• Implementation Steps

- 1) Start point localization
- 2) Real-world motion simulation: After the user's starting point has been created, the navigation system can simulate real-world motion. Taking advantage of the Google AR core, as well as the gyroscope and other sensors on the device, the system can provide an accurate experience as a user by tracking the direction and direction of movement of the user.
- 3) Select Destination: The user selects their desired destination, either through manual input or by interacting with the AR interface.
- 4) A* Shortest Route from Current Location: The NavMesh system uses the A* algorithm to calculate the shortest route from the user's current location to the selected location, as described first. The algorithm evaluates variables based on their values and heuristic predictions (including problems and other relevant factors). When the shortest path is determined, it can be presented to the user visually. This can be done by drawing a line on the screen or presenting an arrow that guides the user along the path to the destination. As picture shown (4)
- 5) Out of bounds recalibration: The system will cause recalibration if the user deviates from the specified path or goes out of bounds. This is done by scanning the QR code as an 8-digit reference or using Vuforia to identify the target image.
- 6) The QR code scanned also provides real-time information about the area.

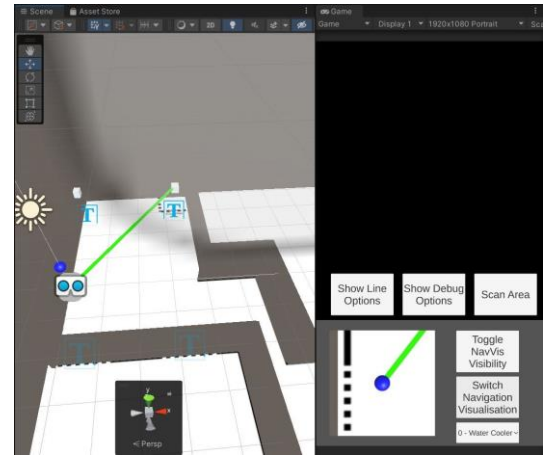


Fig. 5. Development Stage Navigation

C. Smart Appliance control

• Target Image generation and recognition using Vuforia

Vuforia is a powerful platform for target image generation and recognition, which can be leveraged for smart appliance control. In this context, target images representing specific appliances can be created using Vuforia's image target generation tools.



Fig. 6. Example Target image of a light bulb

These target images are then recognized by the Vuforia SDK, allowing the smart appliance control system to identify and interact with the corresponding appliances. By capturing the target image through a device's camera, the system can initiate control commands and enable users to remotely operate and manage their appliances.

• Device control

- 1) Point the camera to scan and detect the target image.
- 2) Respective AR buttons for various commands will be displayed to control the appliance referred to by the target image.
- 3) When the user interacts with these AR buttons, they trigger corresponding actions through the Blynk API. Blynk is a platform that enables communication between hardware devices and mobile apps.
- 4) The Blynk API sends a command over the Wi-Fi network to the ESP32 microcontroller, which is connected

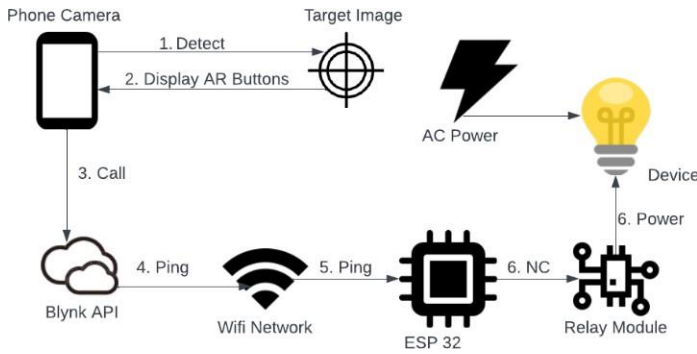


Fig. 7. Working of the Appliance Control System

to the appliances. The ESP32 acts as a bridge between the Blynk API and the physical appliances.

- 5) Upon receiving the command from the Blynk API, the ESP32 activates a specific pin on the connected relay module. The relay module acts as a switch that controls the power supply to the targeted appliance.
- 6) Once the relay module's pin is activated, the power supply to the appliance is switched on. This action results in the device being powered up and ready for use.

D. Real Time updates

- The user scans QR codes placed along the indoor navigation path using the which is interpreted by the ZXing SDK. The ZXing SDK (Zebra Crossing) is a widely used open-source library that enables QR code scanning and decoding functionality in mobile applications.
- Each QR code corresponds to a specific room or location. A JSON file contains the timetable or schedule information for each room. The JSON file includes details such as class timings, events, or any other relevant information.
- The ZXing SDK decodes the scanned QR code, extracting the encoded information embedded within it.
- The decoded QR code data is used to retrieve the corresponding information from the JSON file. This information includes details about ongoing classes and faculty details relevant to the specific room.
- The retrieved information is displayed to the user in real-time.

The ZXing SDK is an open-source library that offers cross-platform compatibility, reliable QR code recognition, and customization options for easy integration into mobile applications. With its widespread usage and active community support, it provides a convenient and reliable solution for implementing QR code scanning functionality.

IV. USABILITY ANALYSIS

We have done the usability evaluation of our system. We provided the users with a smart phone, enabled with our

system and asked them to navigate through the floor, control a light and view the real time information of the activities in the classrooom using the smart phone. Our used smart phone was OnePlus 7t and it has 48 camera, operating system is Android 10, OxygenOS 12.1 and Chipset Qualcomm SM8150 Snapdragon 855+ (7nm). After the users experimental use we asked them some questions.

The questionnaires are given in Table I

	Questions
1	The system is easily adaptable
2	Response rate is Satisfactory
3	Beneficial for disabled
4	Consider using the system in daily life

Based on the user feedback we prepared a graph which based on Likert Scale (see figure 8). From the figure we can see that a large number of people around 89% are agreeing that they can adapt easily with our system. They also agree highly that our system is beneficial for the physically disabled and paralyzed people, also number of people consider to use our system in their daily life.

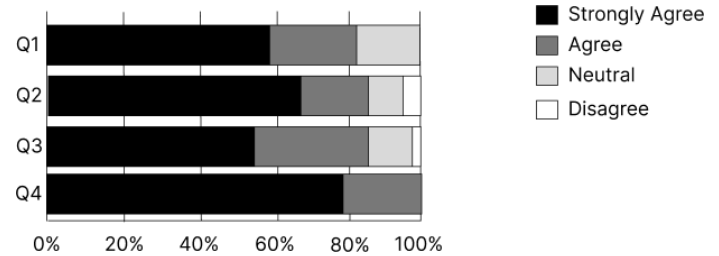


Fig. 8. ESP32 and Relay Module Circuit Diagram

V. SOCIAL IMPACT

The suggested indoor navigation system's integration of augmented reality, real-time information updates, and improved accessibility has a substantial social impact. The system utilises these technologies to meet diverse societal demands and makes numerous positive contributions to people's well-being.

- The suggested indoor navigation system integrates augmented reality, real-time updates, and improved accessibility, greatly impacting society. It meets diverse needs and enhances well-being.
- This method directs users to safety in emergencies. Unlike traditional systems relying on static signage or voice directions, augmented reality offers real-time information, instructions, emergency exits, and evacuation routes. Users navigate challenging indoor environments confidently and efficiently, reducing panic and confusion.
- It supports people with disabilities and allows those with mobility restrictions to operate appliances through the interface.

- The AR interface enables users to adjust lighting, temperature, and other appliances, enhancing their experience and promoting energy conservation. This reduces energy consumption, contributing to environmental sustainability and climate change mitigation.

VI. LIMITATIONS

Although this proposal has many benefits, its limitations must also be acknowledged. The accuracy and reliability of the system depends on the performance of the AR tracker, the environment, and a stable network connection. Rigorous testing and benchmarking to ensure performance in different areas matters. Another limitation is compatibility with different types of devices. Although the system is designed for the control of lighting, fans and other devices with ESP32 and relay modules, some models or brands may experience compatibility problems. In a limited or unreliable network environment, system performance may be affected and users may experience delays or interruptions in receiving updates or control equipment. Mitigation strategies such as caching local data or using offline browsing capabilities can be explored to overcome this limitation. There are also privacy and security concerns to consider. The system collects and processes real-time data, including user positions and control commands. Strong security measures must be taken to protect sensitive data and prevent unauthorized access. User consent and a privacy policy must be established to address privacy concerns and ensure compliance with data protection laws.

VII. CONCLUSION

The proposed augmented reality-based indoor navigation system and appliance control system, utilizing Vuforia, Unity, ZXing QR code scanning, ESP32, relay modules, and the Blynk API, demonstrates significant potential for use in emergency situations, large spaces, and enhanced accessibility and has a profound social impact. By providing real-time information updates, assisting individuals with disabilities, and offering remote appliance control, the system improves public safety, promotes inclusivity, enhances convenience, and contributes to energy efficiency. Its ability to guide users during emergencies, support individuals with disabilities, and foster collaboration with emergency services makes it a valuable tool for creating safer, more accessible, and sustainable indoor environments. By harnessing the power of augmented reality and advanced technologies, the proposed system has the potential to positively transform how individuals navigate and interact with indoor spaces, benefiting society as a whole. In future we could integrate with wearable devices like smart glasses and haptic feedback for hands-free navigation, partner with emergency services for better response and provide a customizable AR interface for improved user experience and accessibility.

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